

From Individual Agents to Social Beings: A Community Network Extension of the Flee3 Framework for the Rohingya Exodus

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Repository: <https://github.com/shaniasinha/rohingya-flee>

As of May 22, 2026, an extended version of this project is still under development.

Abstract—This study extends the *Flee3* agent-based modelling framework to simulate the 2017 Rohingya refugee crisis, focusing on how social and community networks shape patterns of displacement. Building on conflict triggers and demographic distributions, we introduce a religion-based social network extension and a novel metric of *transition probability* to capture sequential movement from IDP camps to Cox’s Bazar. Using empirical data from UNHCR, IOM, ISCG, and ACLED, we compare three configurations: the baseline *Flee3* model, a demographic-informed variant, and a network-augmented variant. Results show that while Cox’s Bazar consistently emerges as the dominant destination across all models, demographic and network extensions provide only marginal improvements in predictive accuracy and clustering realism. Transition probabilities remain stable at approximately 0.178 across configurations, suggesting robustness of aggregate flows to behavioural extensions under short simulation horizons. These findings highlight both the promise and limitations of incorporating social networks into forced migration modelling, with implications for advancing computational migration theory and supporting humanitarian planning.

I. INTRODUCTION

A. Background on the Rohingya Crisis

The Rohingya, a predominantly Muslim minority from Myanmar’s Rakhine State, have long been denied citizenship under the 1982 Citizenship Law and subjected to systemic discrimination and periodic violence [1], [2]. Earlier waves of persecution displaced hundreds of thousands to Bangladesh in 1978, 1991–92, 2012, and 2016 [3]–[6]. The crisis escalated in August 2017, starting with the harrowing August 25 attacks in areas within the Maungdaw and Rathedaung townships. Military “clearance operations” followed by Arakan Rohingya Salvation Army (ARSA) attacks led to mass killings, sexual violence, and the destruction of villages, described by the UN as “a textbook example of ‘ethnic

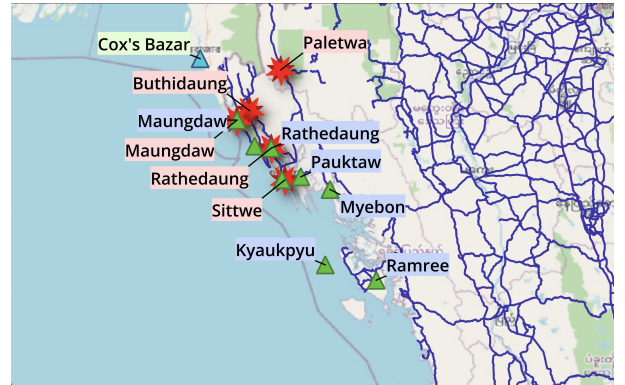


Fig. 1: Map of Rakhine State, Myanmar, showing conflict events (red stars), IDP camp locations (green triangles), and the primary refugee destination in Bangladesh, Cox’s Bazar (blue triangle), used as inputs for the displacement model. Myanmar road network is shown in blue.

cleansing” [7]. Within three months, over 700,000 fled to Bangladesh, with rapid settlement in Cox’s Bazar [8]. By early 2018, more than one million Rohingya lived in Bangladesh, forming the world’s largest refugee camp complex [9].

B. Why This Case Matters for Modelling

The crisis also illustrates the importance of social and familial networks in shaping refugee and internally displaced person (IDP) flows. Displaced persons are considered to hold refugee status if they have been forced out of the country due to extenuating circumstances. Some persons in these conflict zones may, however, remain displaced with country borders and are considered to be IDPs [10]. Anecdotal and survey evidence suggests that Rohingya often fled in extended family groups and sought refuge in areas where relatives or community members from earlier displacement waves were already present [11]. Migration studies more

broadly have demonstrated that such networks act as powerful “pull factors,” often outweighing economic or geographic considerations [12]. Testing the hypothesis that social ties influenced the distribution of Rohingya refugees is therefore both academically significant and practically relevant.

Finally, modelling such a crisis has humanitarian value. Aid agencies faced enormous challenges in scaling up resources to match the speed and volume of refugee and IDP arrivals in late 2017 [13]. Predictive modelling could have improved contingency planning by anticipating arrival surges and likely camp concentrations, allowing for better allocation of food, shelter, and medical supplies. Retrospective modelling of this case can therefore inform the development of forecasting tools for future crises.

C. Relevance to Computational Social Science

Forced migration has become a significant focus in computational social science, with models ranging from gravity-based approaches to agent-based simulations [14], [15]. While these models have been applied to crises in Syria, South Sudan, and elsewhere, relatively few studies explicitly account for the role of social networks in refugee decision-making. This study thus aims to fill a critical gap: it combines strong theoretical expectations about the influence of family and community ties with empirical data against which models can be tested.

This paper makes three contributions:

- 1) We apply the *Flee3* ABM framework to the 2017 Rohingya crisis, compiling and using available conflict and displacement data.
- 2) We extend the model to incorporate network-informed decision rules, where agents’ choices of destination are shaped not only by geography but also by the presence of community ties due to religion.
- 3) We evaluate the effect of these modifications on predictive accuracy, with implications for both humanitarian planning and computational migration theory.

By situating the Rohingya crisis at the intersection of empirical richness and theoretical significance, this study demonstrates how computational modelling can both enhance academic understanding of migration dynamics and contribute to improving real-world humanitarian response.

II. RELATED WORK

Research on forced migration spans multiple disciplines, from political science and demography to com-

putational modelling. In this section, we review three strands of work most relevant to this study: (i) modelling approaches to refugee and IDP flows, (ii) applications of the *Flee3* framework and related agent-based models, and (iii) migration theory emphasizing the role of social and familial networks.

A. Modelling Refugee and IDP Flows

A wide range of approaches have been developed to model displacement under conflict. Macro-level statistical models, such as gravity or push-pull frameworks, estimate flows based on geographic, economic, and political variables [16], [17]. These models capture large-scale correlations but often lack micro-level behavioural realism.

In contrast, agent-based models (ABMs) simulate the decisions of individual or household agents, allowing researchers to incorporate heterogeneity, network effects, and non-linear dynamics [18]. ABMs have been increasingly used to study forced displacement because they can flexibly encode movement rules tied to conflict events, road networks, or social interactions. Their fine-grained representation makes them particularly useful for evaluating camp-level distributions and short-term flows, where macro-level models struggle.

B. The *Flee3* Framework

Among ABMs for refugee modelling, *Flee3* [15] has emerged as a widely used, open-source framework. Designed for rapid deployment in humanitarian contexts, *Flee3* represents refugees as agents moving across a simplified geographic network, where towns and camps are nodes and travel routes are edges. Conflict events are encoded as triggers that prompt agent departure, and flows are calibrated against UNHCR arrival data.

The *Flee3* framework has been applied to crises in South Sudan, Syria, and the Democratic Republic of Congo, demonstrating its ability to reproduce broad spatio-temporal displacement patterns [14], [15], [19]. Extensions of *Flee3* have explored uncertainty quantification, hybrid modelling with differential equations, and integration with high-performance computing platforms for real-time forecasting [19].

Despite these successes, *Flee3* and similar ABMs often rely primarily on geographic shortest-path assumptions, with limited representation of the social processes that guide refugee decisions. This creates an opportunity to test whether incorporating social network dynamics — such as kinship ties or prior migration corridors — can improve predictive accuracy.

C. Social Networks in Migration Theory

Social and familial networks have long been recognized as central to migration theory. *Network migration theory* posits that migration decisions are shaped by prior migrants who lower informational and economic barriers for subsequent movers [20]. In refugee contexts, such networks often determine not just whether individuals flee, but where they go. Empirical studies have shown that refugees tend to cluster where relatives or co-ethnics are already present, reinforcing patterns of community-based settlement [11], [21].

In the case of the Rohingya, such dynamics are particularly salient. Earlier displacement waves in 1978, 1991–92, and 2016 created established Rohingya communities in Bangladesh, especially in Cox’s Bazar. Anecdotal and survey evidence suggests that during the 2017 exodus, many families aimed for locations where relatives were already settled, reflecting strong kinship and religious cohesion [6], [8].

Yet, relatively few computational models of displacement explicitly incorporate such social factors. Integrating network-driven decision rules into frameworks like *Flee3* offers a promising avenue to bridge migration theory with computational practice, and to assess empirically whether such extensions improve predictive realism.

III. THEORY

A. Modelling Framework

The simulations in this study are conducted using *Flee3*, an open-source agent-based modelling framework for forced displacement [15] and *FabFlee*. Individuals are represented as autonomous agents navigating a simplified geographic graph composed of locations and routes. Locations correspond to towns, villages, or refugee camps, while routes represent traversable connections such as roads or paths. The routes for this model use existing connections between two locations from the actual road network, and connect the two locations using a straight line.

At each discrete time step, agents evaluate possible destinations and update their positions accordingly. Displacement is triggered by the occurrence of conflict events, which cause populations to leave unsafe locations and move toward camps. Agents select routes and destinations on the basis of decision rules that weigh distance, safety, and capacity. Because of its modularity and efficiency, *Flee3* and *FabFlee* provide a robust platform for testing behavioural extensions.

B. Data Sources and Availability

Data on the 2017 Rohingya crisis are asymmetrical. Refugee flows into Bangladesh are well documented through UNHCR, IOM, and ISCG arrival counts, camp mapping, and registration data. In contrast, information from Myanmar is limited: the Rohingya were excluded from the 2014 census [22], access to Rakhine was restricted, and conflict events were extracted from partially incomplete datasets such as ACLED. To account for the census exclusion, we adjusted demographic distributions to approximate the true religious composition of the Rakhine state and union, encoding the transformed data in `demographics_religion.csv`. The assumption made is that the places with the most conflicts starting August 25, 2017 (Rathedaung, Maungdaw and Buthidaung) had a higher Muslim population, since the attacks were targeted towards the Rohingya. These locations have a higher Muslim population (70-80%) compared to the default value of around 35%. Cox’s Bazar also has a higher percentage of Muslim population since it already had many Rohingya refugees prior to the August 25 attacks, and Bangladesh is a Muslim-majority country.

C. Religion-Based Social Network Extension

While the default *Flee3* framework bases agent decisions on geographic and safety considerations, empirical research suggests that displaced populations often rely heavily on social networks when selecting destinations [12]. To capture this effect, we introduce an extension that allows agents to form and act upon social ties defined by shared religion. A reason to choose religion-based network was that there was sufficient data available that took the Rohingya population into account. Other attributes such as ethnicity, language and family and social ties could be considered, but not enough data was found for the time period being modelled here.

Each agent is assigned a religion attribute at initialization using an external demographics file (`demographics_religion.csv`), and is then probabilistically linked to a bounded number of other agents of the same religion. This produces a network structure characterized by homophily and limited degree, reflecting realistic patterns of social clustering.

During movement decisions, the social network contributes an additional utility term to the destination evaluation function. The extended utility is defined as:

$$U(d) = \alpha D(d) + \beta C(d) + \gamma S(d) + \delta N(d) \quad (1)$$

In Equation 1, $D(d)$ represents distance-based utility, $C(d)$ represents capacity-based utility, $S(d)$ reflects

safety-based utility, and $N(d)$ denotes the social network-based utility. α, β, γ and δ are configurable weight parameters. $N(d)$ is measured as shown below in Equation 2,

$$N(d) = \frac{|C_{\text{agent}} \cap A_d|}{|C_{\text{agent}}|} \quad (2)$$

where C_{agent} is the set of social connections for the agent, A_d denotes the set of agents currently at destination d , and $|C_{\text{agent}} \cap A_d|$ denotes the number of agent's connections present at destination d .

The parameter δ regulates the relative importance of social ties in comparison to geographic and safety factors. In practice, when agents evaluate potential destinations, they are drawn not only toward locations that are close and secure, but also toward those where members of their religious network are already established.

This design is fully modular: the social component can be enabled or disabled through configuration in `simsettings.yml`, and the maximum number of links, link probability, and network weight can be tuned without altering the base model.

D. Transitioning from IDP to Refugee

To capture sequential movement dynamics between temporary and long-term destinations, we introduce a custom metric that we call the *transition probability*. This is not a standard quantity in migration modelling but is defined here to quantify the proportion of agents who first entered an IDP camp before subsequently arriving in Cox's Bazar:

$$P_{\text{transition}} = \frac{N_{\text{IDP} \rightarrow \text{Ref}}}{N_{\text{total}}}, \quad (3)$$

where $N_{\text{IDP} \rightarrow \text{Ref}}$ is the number of agents following the IDP-to-Refugee trajectory and N_{total} is the total number of agents spawned in the simulation.

To measure consistency across simulation runs, we treat $P_{\text{transition}}$ as a random variable and calculate variance and standard deviation:

$$\sigma^2 = \frac{1}{R} \sum_{i=1}^R (P_i - \bar{P})^2, \quad (4)$$

$$\sigma = \sqrt{\sigma^2}, \quad (5)$$

where P_i is the transition probability for run i , $\bar{P}$ is the mean across runs, and R is the number of runs. These statistics provide a measure of stability and allow comparison between different model configurations.

IV. METHODS

A. Implementation Details

1) *Data Structure Modifications*: To incorporate social network functionality into the *Flee3* framework, modifications were made at both the agent and system levels.

At the agent level, the *Person* class was extended with three new attributes: a unique agent identifier (`agent_id`), a list of social links (`social_links`), and a cached set representation of these links (`_social_links_set`). The identifier allows efficient referencing of agents across the network, while the adjacency list records each agent's social connections. The cached set ensures constant-time lookup operations during movement decisions, thereby reducing computational overhead.

At the ecosystem level, additional infrastructure was introduced to manage and query social networks. A global counter (`next_agent_id`) was implemented to assign unique identifiers to newly spawned agents. Two dictionaries were added: one mapping demographic groups to their respective agents (`agents_by_demographic`) and another mapping identifiers directly to agents (`agents_by_id`). This design enables efficient access both by religion (for network formation) and by ID (for movement calculations).

2) *Social Network Formation Algorithm*: Social connections between agents are generated at spawn time using a probabilistic algorithm designed to produce realistic network topologies. Each new agent is linked to others of the same religion with a probability defined by the parameter `SocialNetworkLinkProbability`, subject to a maximum number of links per agent (`SocialNetworkMaxLinks`). This process results in sparse, bounded-degree networks that reflect homophily while preventing unrealistic super-connected individuals. All links are bidirectional, ensuring mutual recognition of ties.

Four parameters govern network formation, summarized in Table I. These allow flexible calibration of social influence while maintaining computational efficiency. The parameters used for the network model are given in Table II.

3) *Integration with Movement Decisions*: The social network influences agent movement through an additional "social attraction" term in the destination scoring function. For a given agent, the proportion of its social connections already present at a potential destination is computed as the network utility. This value is then scaled by `SocialNetworkWeight` and added to the existing

TABLE I: Configurable parameters for social network formation in `simsettings.yml`.

Parameter	Range / Values
SocialNetworkEffect	Boolean {True, False}
SocialNetworkAttribute	"religion", "ethnicity". etc.
SocialNetworkWeight	Float [0.0 – 1.0]
SocialNetworkMaxLinks	Integer [1 – k], $k \in \mathbb{N}$
SocialNetworkLinkProbability	Float [0.0 – 1.0]

TABLE II: Chosen parameters for model with network based on religion.

Parameter	Value
SocialNetworkEffect	True
SocialNetworkAttribute	"religion"
SocialNetworkWeight	0.5
SocialNetworkMaxLinks	5
SocialNetworkLinkProbability	0.8

destination score, which otherwise accounts for distance, safety, and capacity.

This integration was implemented within the `getEndPointScore` function in *Flee3*. If `SocialNetworkEffect` is enabled and an agent has established links, the network utility is calculated and incorporated into the score. By doing so, destinations with higher concentrations of an agent’s social network become more attractive, increasing the likelihood of agents clustering with co-religionists or kin. This design preserves backward compatibility: when the feature is disabled, the framework defaults to its original behaviour.

B. Computational Considerations

1) *Time and Space Complexity*: Network formation requires at most $\min(k, n)$ operations per agent, where k is the maximum number of links and n is the number of co-religionists. For a population of size N , the total cost of link formation is therefore $O(N \times \min(k, n))$, with space complexity $O(N \times k)$.

Movement decisions scale with the number of destinations d . Without optimization, score evaluation requires $O(m \times a)$ per agent, where m denotes the number of agents already at the destination and a the number of social links held by the agent. With set-based caching, this reduces to $O(m)$ per agent.

Hence the overall cost per time-step is:

$$O(N \times D \times m).$$

Taken together, the overall simulation complexity becomes:

$$O(T \times N \times D \times m + N \times \min(k, n)),$$

compared to the baseline implementation of $O(T \times N \times D)$ in *Flee3*. Appendix A discusses the actual computation time taken by the specific machine we used to run 10 runs of three variations of the conflict scenario, including this network-version.

2) *Optimizations*: To mitigate the additional overhead, several optimizations were introduced. Storing social links as sets reduces membership queries from $O(k)$ to $O(1)$. Conditional checks (“early exits”) avoid unnecessary computation when the feature is disabled, when agents lack connections, or when destinations are empty. Finally, lazy evaluation ensures that cached sets are only created when required, reducing memory use for agents without active social interactions.

3) *Scalability*: Since both k and m are bounded, the extension preserves linear growth in N . Sparse networks, where the probability of forming links is small ($p \ll 1$), impose negligible overhead. Dense networks, where the probability approaches one ($p \rightarrow 1$), increase memory usage, but this growth remains bounded by $O(N \times k)$. As a result, the extension scales efficiently while maintaining realistic clustering behaviour.

C. IDP-to-Refugee Transition Probability

To evaluate how agents transitioned between intermediate displacement sites and the main refugee destination, we developed a Python script to process simulation output across three model configurations (found in the repository under `myanmar2017`, `myanmar2017_demo`, and `myanmar2017_network` respectively for the base, demographically-informed, and network-extended models). The script was optimized for large-scale data (approximately five million agent records) by chunked processing, corrected for naming inconsistencies in the output files, and generalized to analyze all agents regardless of religion. For each run, the script identified agents that first entered an IDP camp and later arrived in Cox’s Bazar, then aggregated results across 10 independent runs for each configuration. The outputs included both detailed run-level results and summary files reporting mean values, variances, and standard deviations of transition probabilities.

V. RESULTS & DISCUSSION

A. Validation versus Simulated Data

Heatmaps of mean error (across 10 different runs of each model) between simulated and observed agent flows for three model configurations are shown: Figure 2a shows the baseline *Flee3* implementation (`myanmar2017`), Figure 2b shows a demographic-informed variant (`myanmar2017_demo`), and Figure 2c

shows the mean errors network-augmented variant (myanmar2017_network). The errors represent the difference between the validation data and simulated data on agent flow, as calculated by the existing *Flee3* code. Temporal accuracy can be observed by tracking deviations across dates and spatial accuracy can be observed by comparing errors across locations. Each row is scaled between 0 and 1 to allow for comparing accuracy across camps, since they have very different population scales.

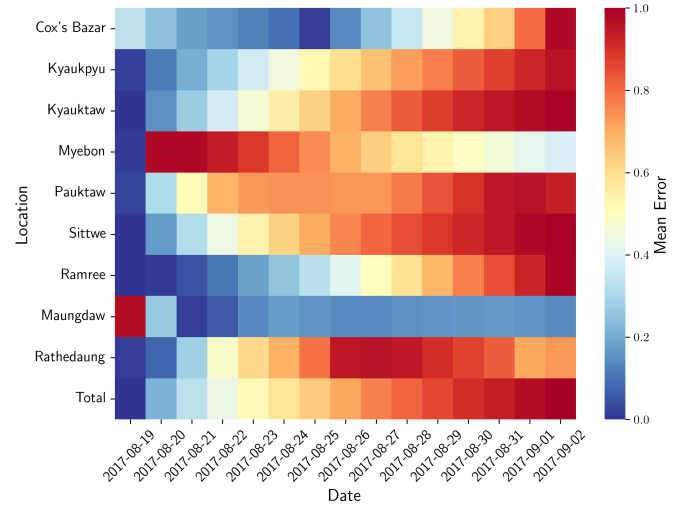
No obvious differences are observed across the three models. Cox’s Bazar shows steadily increasing error toward the later part of the simulation, while Rakhine townships such as Maungdaw and Rathedaung exhibit fluctuating mismatches between observed and simulated outflows. Maungdaw and Myebon are the only two locations for which the error reduces as the simulation continues. For Myebon, the error reduces in the post-conflict period (post 25th August 2017). For all the IDP camps, the lowest errors were observed before the conflicts began. For Cox’s Bazar, the lowest error was observed around the peak of the conflict.

These results suggest that changing the simulation configurations and adding a network-based pull factor does not significantly contribute to making the model more accurate.

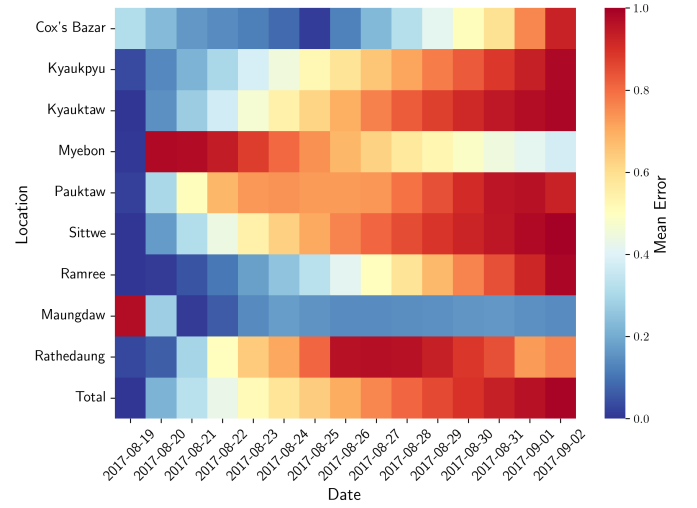
B. Tracking Displaced Rohingya across Camps

Figure 4 compares the simulated Muslim agent populations over time across major camps and townships for the demographic-informed (left) and network-augmented (right) variants of the myanmar2017 model. The y-axis indicates the number of Muslim agents, interpreted as a proxy for Rohingya displacement. In both configurations, Cox’s Bazar emerges as the dominant destination, reflecting its proximity to the border and its role as the primary reception site during the 2017 exodus. Peripheral sites such as Myebon or Ramree attract relatively few agents. The close similarity of the two subplots suggests that the addition of network effects does not substantially alter aggregate population trajectories over the simulated period.

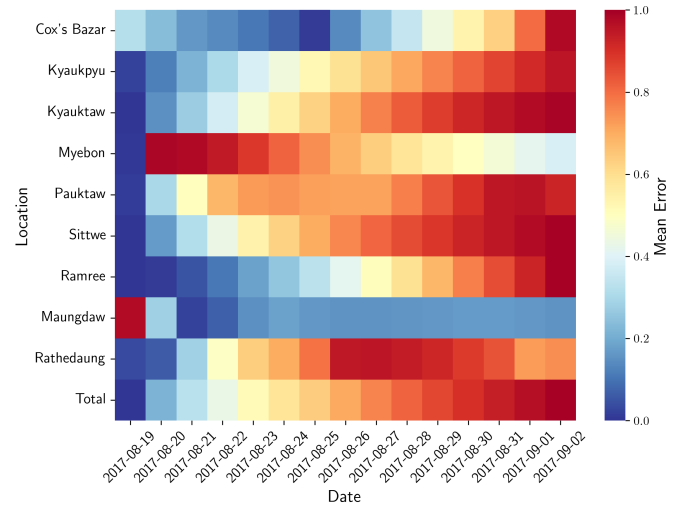
Because the Myanmar census and official data excluded the Rohingya, direct population counts are not available. In this model, Muslim agents are therefore used as a proxy for Rohingya agents, with demographic adjustments applied to approximate their true distribution in Rakhine State. This makes the graph particularly valuable: it highlights how displacement patterns unfold across camps and townships despite the absence of reliable Rohingya-specific census data, thereby providing a more faithful representation of the 2017 exodus.



(a) Model: myanmar2017 model



(b) Model: myanmar2017_demo



(c) Model: myanmar2017_network

Fig. 2: Comparison of heatmaps across three models displaying agent flow prediction error. Note that each row of each heatmap is scaled between 0 and 1.

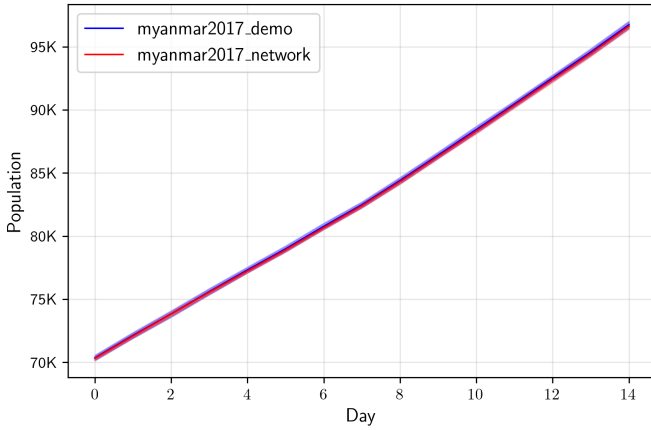


Fig. 3: Muslim agent arrivals in Cox’s Bazar under the demographic-informed (blue) and network-augmented (red) models, showing nearly identical trajectories with only minor differences.

To assess whether network effects influence aggregate flows into the primary refugee destination, we compared the number of Muslim agents arriving in Cox’s Bazar under the demographic-informed and network-augmented variants. Figure 3 shows that both models produce near-linear growth in arrivals over the two-week simulation period, with trajectories closely aligned. The demographic-only model yields slightly higher estimates of arrivals, particularly in the later stages of the run, but the magnitude of this difference remains small relative to the total population size. This indicates that the incorporation of network effects does not significantly alter aggregate flows into Cox’s Bazar. These flows are primarily driven by its geographic proximity to northern Rakhine and its function as the principal reception site. Agent flows into secondary destinations are not significantly impacted either, as shown in Figure 4.

C. IDP-to-Refugee Transition probability

Figure 5 reports the mean transition probability, defined as the proportion of agents who first entered an IDP camp before subsequently arriving in Cox’s Bazar, across the three model configurations. The shaded region denotes one standard deviation across 10 independent runs.

All three variants yield very similar outcomes, with mean transition probabilities clustered around 0.178. The demographic-informed model (`myanmar2017_demo`) produces the highest value (0.1781), while the network-augmented model (`myanmar2017_network`) produces the lowest (0.1774). Standard deviations are extremely small (on the order of 10^{-3}), indicating consistent results across runs.

The narrow range of variation suggests that neither demographic calibration nor network extensions substantially alter the likelihood that agents transition through IDP camps before reaching Cox’s Bazar. This stability underscores that, under the present assumptions and parametrizations, aggregate displacement pathways are robust to the inclusion of additional behavioural mechanisms.

It should be noted that the transition probability, as defined in this study, is not a standard measure in forced migration modelling but rather a construct introduced to capture sequential movement dynamics between intermediate and final destinations. While ad hoc, this metric provides a useful lens for comparing model variants and assessing whether behavioural extensions influence multi-stage displacement trajectories.

VI. FUTURE WORK

Several limitations of the current study point toward valuable directions for future research. First, validation is constrained by limited empirical data on Rohingya populations and attributes during the 2017 crisis. This scarcity may explain why the network-augmented variant did not outperform other models. Future studies should seek finer-grained data such as detailed movement timelines or community-level surveys to enable more rigorous model validation.

Second, our simulation evaluates only short-term displacement, whereas longer timelines may better reveal the impact of social networks. Empirical studies have shown that short-term migration often responds to urgent safety and proximity incentives, while long-term migration is more responsive to social ties (e.g., [23], [24]). Extending the simulation period could thus reveal network effects that do not manifest immediately.

Third, the current network construction relies solely on religion. However, Rohingya identities and migration behaviours are shaped by multiple overlapping ties, including kinship, ethnicity, and shared community. Future work should explore richer, multidimensional network models to better capture clustering and displacement patterns, especially over longer periods.

Finally, the conflict model currently uses binary indicators, treating all events as equivalent. A more nuanced approach would assign graded conflict intensities, potentially derived from ACLED data by scoring event severity or sentiment. Automating such conflict scoring would improve realism and could more accurately represent varying levels of threat or access constraints.

Implementing these enhancements—improved validation, longer simulation horizons, multidimensional social

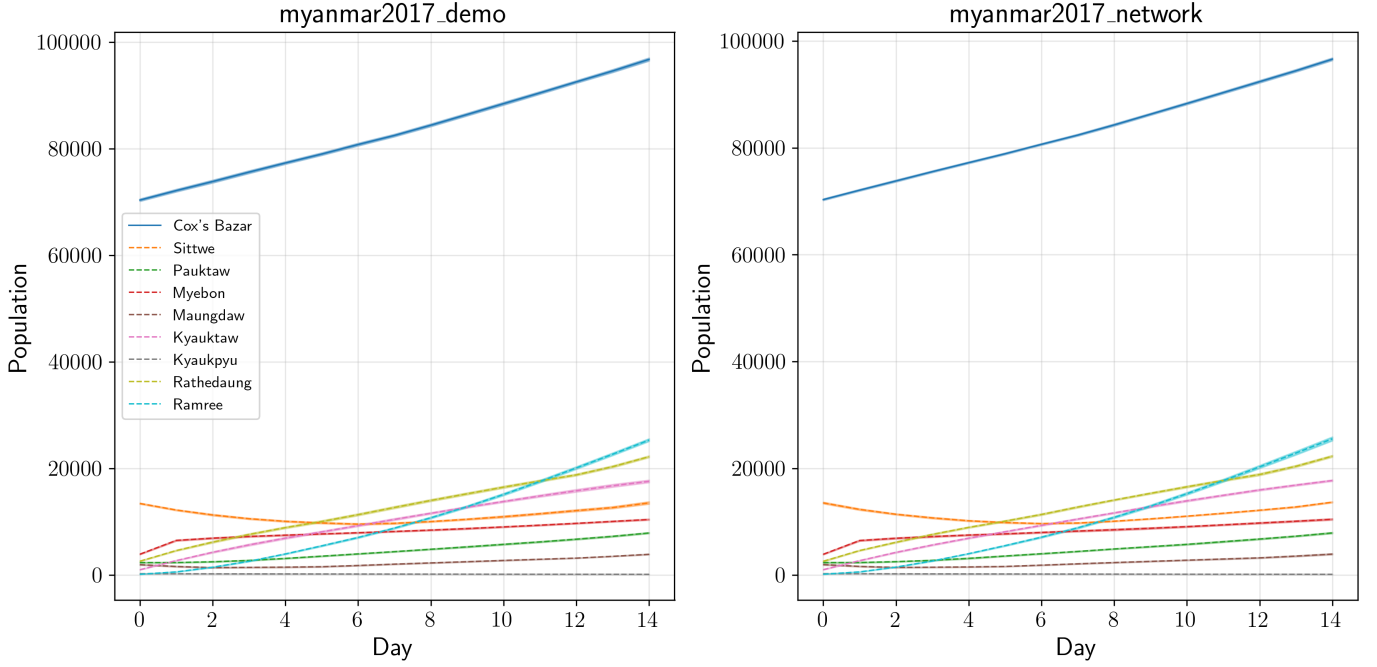


Fig. 4: Simulated Muslim agent populations for the demographic (myanmar2017_demo, left) and network (myanmar2017_network, right) models, using Muslim agents as a proxy for Rohingya displacement.

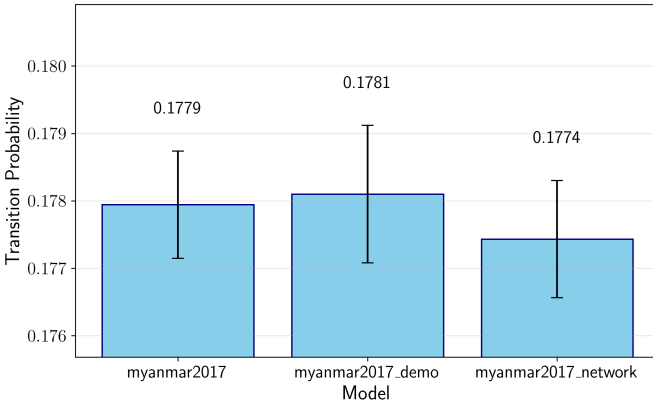


Fig. 5: Mean transition probability of going from an IDP camp to a refugee camp, with standard deviation across 10 runs for each model variant.

network design, and refined conflict representation—will improve Flee3’s ability to simulate complex forced displacement dynamics with greater accuracy and relevance.

VII. CONCLUSION

This project has implemented and evaluated a community network extension of the *Flee3* framework to model the displacement of Rohingya refugees during the August 2017 crisis. By integrating religion-based networks and a custom transition probability metric, the study tested whether social ties alter aggregate migration trajectories.

The results indicate that although the demographic and network-augmented models marginally reduce error in certain locations, overall flows remain dominated by geography and conflict triggers, with little divergence from the baseline. Transition probabilities are stable across model variants, underscoring the robustness of aggregate flows but also suggesting that short-term displacement may be less sensitive to social influences than longer-term processes.

The work demonstrates both the feasibility and computational trade-offs of embedding social mechanisms into ABMs of forced migration. While the present extensions did not produce substantial performance gains in predictive power over the base models, they lay the foundation for richer modelling of kinship, ethnicity, and other forms of community ties in future studies. Improved empirical validation, longer simulation horizons, and more nuanced conflict intensity measures will be critical to realizing the potential of network-informed refugee modelling. In doing so, this line of research can deepen theoretical understanding of displacement dynamics while also contributing practical tools for humanitarian forecasting and response.

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APPENDIX A

COMPUTATION TIME OF MODELS

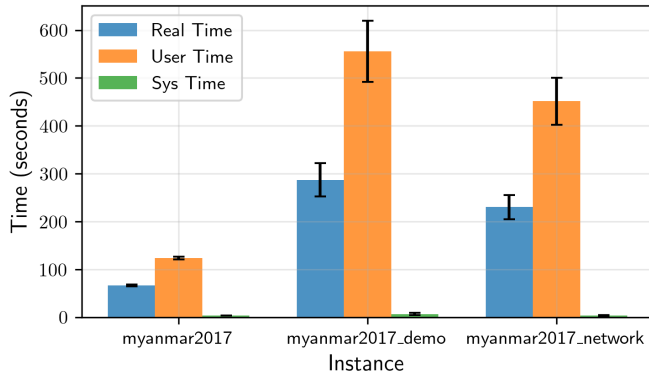


Fig. 6: Average computation time (real, user, and system) across 10 runs for baseline, demographic, and network models.

Figure 6 compares the computational costs of the three model variants in terms of real time, user CPU time, and system time, averaged across 10 runs. Computational performance was assessed by instrumenting each simulation run with automated timing collection. For each model variant, we recorded *real time* (wall-clock duration), *user time* (CPU time spent in user mode), and *system time* (CPU time spent in kernel mode). Each configuration was executed 10 times, and the mean and standard deviation of these measurements were reported. All experiments were conducted on a MacBook Pro 2019, equipped with a 2.4 GHz Intel Core i5 with 4 cores, 8 GB RAM and running MacOS Sequoia 15.4.1.

The baseline configuration (myanmar2017) executes in under 2 minutes of real time, while both the demographic-informed and network-augmented variants require significantly more resources. The demographic model is the most expensive, averaging nearly 300 seconds of real time and over 550 seconds of user CPU time. The network-augmented model is somewhat more efficient but still incurs a substantial increase relative to the baseline. System time remains negligible in all cases, indicating that performance is primarily bounded by user-level computations rather than I/O overhead.

The higher cost of the demographic-informed model likely arises from overhead during agent initialization and demographic assignment, which requires repeated lookups and updates to population-specific data structures. By contrast, the network extension primarily affects movement decisions, where computational load is mitigated by bounded connectivity and set-based optimizations. As a result, the demographic variant is slower than the network-augmented one despite its simpler conceptual design.

These results highlight the trade-off between model realism and computational efficiency. While demographic and network extensions improve behavioural fidelity, they substantially increase runtime, which becomes critical when scaling to large agent populations or conducting extensive parameter sweeps. Reporting computational performance highlights the practical feasibility of running such models in real-world research settings, where computational budgets and time constraints often limit the scope of experiments.